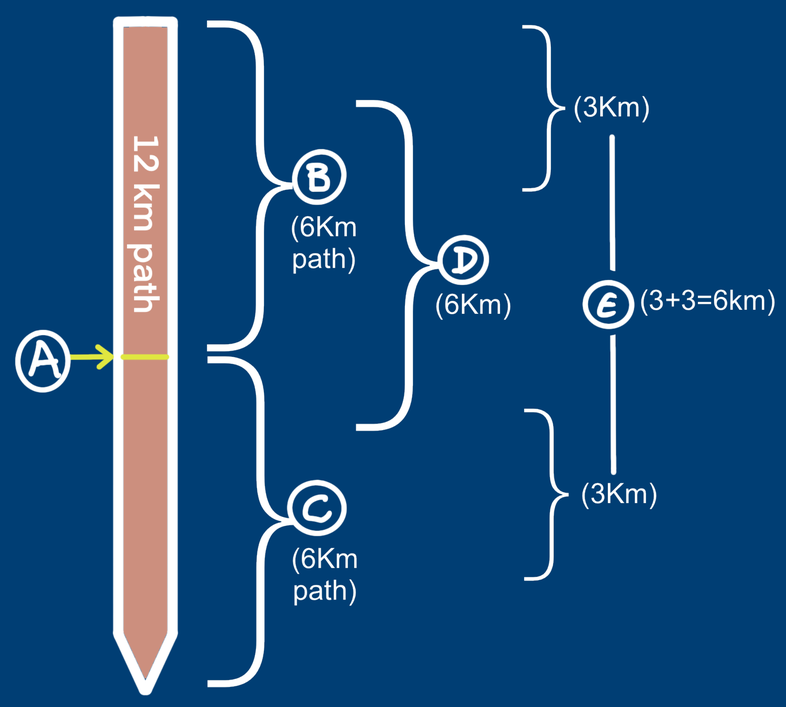


- When an object is divided in two equal parts, a number of potential points of reference are created: the two parts themselves (Ⓑ, Ⓒ, i.e. the "halves"); their contact point Ⓐ; a measure calculated on the size of the halves Ⓓ, or even a collection of subparts whose total is ½ the original Ⓔ.
- Different languages lexicalize these meanings in different ways (English *half/mid*, Italian *metà/mezzo*, Dutch *helft/half*, German *halbe/Hälfte*), with non-trivial mappings.
- These expressions have unusual linguistic properties (bareness, definiteness, semi-optional partitivity).
- Their behavior can be accounted for in terms of:
  - independent assumptions about how DPs can be licensed via silent indefinite Dets, (weak) definites Dets or N-to-D raising;
  - the idea that some *half*-words combine productively only with Ns that have a 'linear dimension' ('path' but not 'love'), and that the plurality denotated by a DP can be 'linearized' by a nominal operator in partitive cases.

Splitting 'half'-words meanings in Italian and English

Michelangelo Falco & Roberto Zamparelli

English data: *half*

Half combines with a partitive *of*-PP (*of* often omitted) followed by a plural, singular, or mass DP (1a).
Half in (1b) may apply to the plurality/amount as whole (fig 1A) or distribute to the atoms (fig.1B).

(1) a. I want half of her confidence / love!
b. Half of the flowers is underwater.
c. It took half a century.

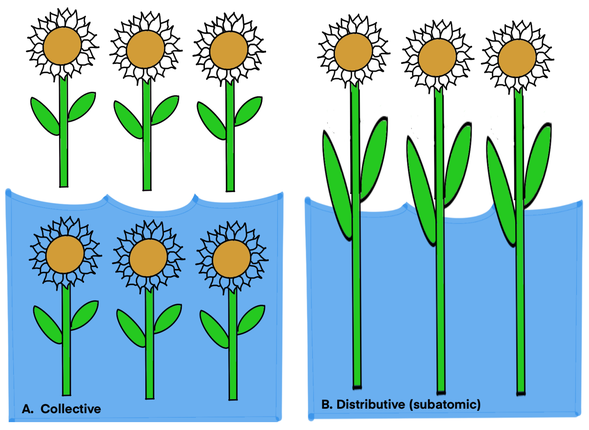


Fig. 1: "Half of the flowers are underwater"

Half can also take N directly, with many idiomatic cases (*half life/brother*...), but productively with Ns that can appear in *the length/duration of a N* (2).

(2) Half {hour, year, century, way, mile, marathon, dozen, day, dollar, inch, goal, term, month, truth, point, time, space...}

Crucially, *half* doesn't combine directly with many gradable Ns that are fine with PPs, but do not have a natural length: (2a) vs. **I want half confidence/love*.

Q: Why can *half*+PP apply to Ns that it would not take directly?

Proposal
In partitives, *half* applies to an operator SIZE, which converts a DP plurality into a corresponding length. See the tree in (12).

In the figure in the top bar, *half* can refer to the Ⓑ, Ⓒ intervals.
But it cannot refer to Ⓐ (for which *mid-* is used) see (3a), nor to the pure measure Ⓓ, see (3b).
Half can appear under Dets, even with the nouns that it rejects when it is bare see *confidence* in (3c).

(3) a. *He stopped at/in half (of the) journey.
b. *The table is half of the sofa long.
c. her half-confidence / two half books

Italian data: *metà* and *mezzo* and their mappings

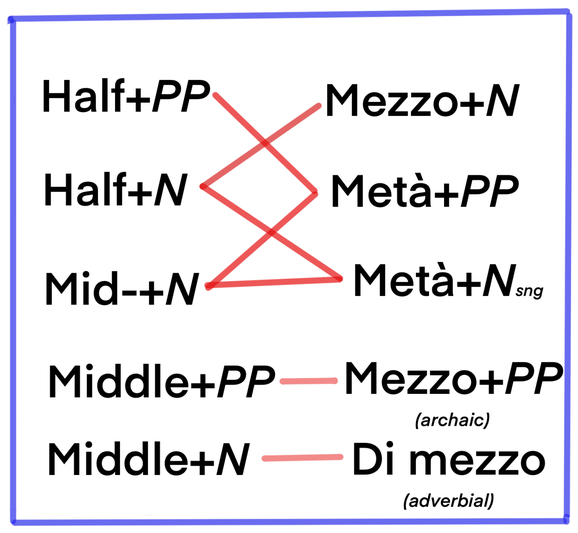


Fig. 2: the *Half* family

Both *metà* and *mezzo* may appear bare or with a determiner.
Metà+N (with restrictions similar to English (2)) is obligatorily bare (4).
Metà di+PP closely corresponds to (1), but the distributive meaning prefers a (definite) determiner (see *??(la) metà (inferiore) dei fiori va sott'acqua* '(the) half of the flowers goes underwater', fig 1B).
Mezzo+N, in (5), corresponds to (3c), and can apply to more abstract meaning components (6).

(4) Farò (*la) metà {percorso / *percorsi}.
I_will_do (the) half {path / paths}

(5) {(Un) mezzo percorso / (dei) mezzi percorsi}.
I{(a) half path / (some) half paths}

(6) Lui è {un mezzo genio / ?? metà di un genio}
he is {a half genius / half of a genius}

(In)Definiteness

Bare *half/metà* have an indefinite meaning (see (7a) vs. **The flowers are here and the flowers are there*) which does not license relatives (7b) and does not allow bare plurals (7c).

(7) a. Half of the flowers are here, half there.
b. *Half of Rome that is East of the Tiber.
c. ??Halves of the notebook were reconnected.

In *half/metà*+N cases, N behaves as a singular *definite*, requiring a singular antecedent.

(8) Ho "Vertigo", ma metà film è rovinata.
I_have "Vertigo", but half movie is ruined.
'... half of **the** movie (= Vertigo)...'

(9) *Ho due film, ma metà film è rovinata.
I_have two movies, but half movie is ruined

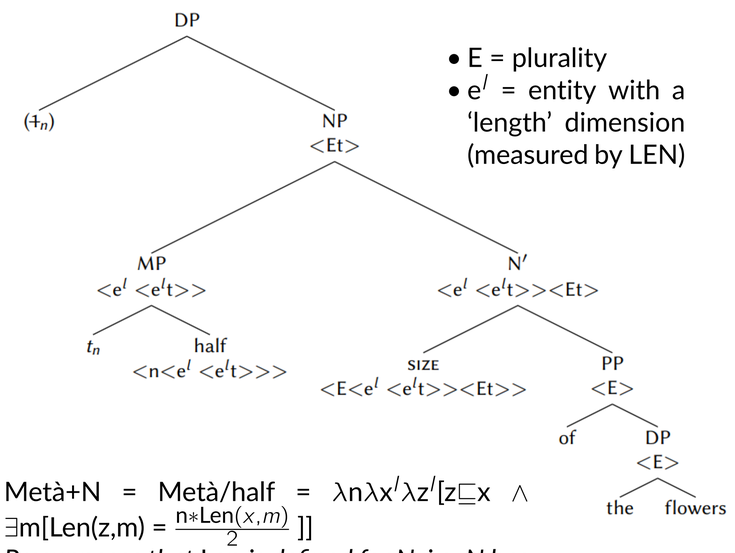
Q: Why bare singulars? (cf. **Third of the flowers*)
Q: Which is the source of the (in)definiteness pattern?

Proposal:
When a specific numeral is forced, it can be left unpronounced, yet semantically active enough to license the (apparent) bare *half/metà*+PP, giving it an indefinite meaning.
(10) $[_{DP/NumP}\ 1\ [_{NP}$ half/metà $[_{PP}$ of the population]]]

Proposal:
Definiteness comes from movement of the operator *metà/half*+N to [Spec,DP] (cf. bare noun coordinations like *Cat and dog were fighting*, Heycock&Zamparelli, 2002)
(11) $[_{DP}\ [_{NP}$ metà N] $\ldots\ t_i]$

Semantics

(12) Semantic-type tree for *half*+PP (collective reading)



(13) Metà+N = Metà/half = $\lambda n \lambda x^l \lambda z^l [z \sqsubseteq x \wedge \exists m[\text{Len}(z,m) = \frac{n * \text{Len}(x,m)}{2}]]$
Presupposes that Len *is defined for N, i.e. N has 'length'*

(14) SIZE = $\lambda x \lambda f \lambda y [y \subseteq x \wedge \exists m \exists z^l [\text{Len}(z,m) = \text{AMOUNT}(x,M) \wedge \forall k[f(z)(k) \rightarrow \text{Len}(k,m) = \text{AMOUNT}(y,M)]]]$

Where AMOUNT(x,M) gives the cardinality if x is a plurality, a mass measured along some contextual unit of measure M if x is a mass or a mereological atom, e.g. *half of the apple*; this obtains discontinuous subparts like Ⓔ (see Wągiel, 2021).

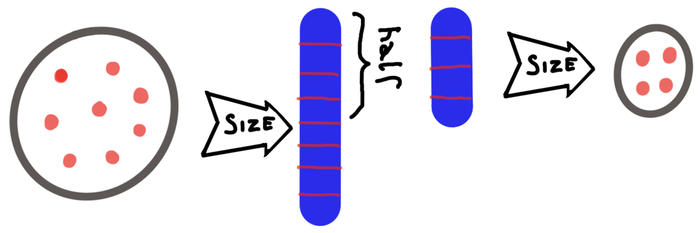


Fig. 3: The functioning of the SIZE operator in the plural case

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References:

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